Vidyalankar Institute of Technology
Semester V – INFT - Mid Semester Assessment – I

Date: 04/08/2025	Machine Learning	PC_PCC	20 Marks/ 45 Minutes

1	**Solve any one (10 marks each)**		**CO**
	A	What is Machine Learning? Explain the different types of machine learning algorithms. For each type, discuss any two real-life applications.	CO1
	B	A company is analyzing customer purchasing behavior using a dataset with two features: Customer | Feature X_1 (Spending on Electronics) | Feature X_2 (Spending on Clothing) A | 4 | 2 B | 2 | 3 C | 3 | 4 D | 5 | 6 Solve first Principal Component and corresponding data transformation values. Use proper formulas at each step to show calculations.	CO1
2	**Solve any one (10 marks each)**		
	A	Answer the following: - (a) Explain how different Dimensionality Reduction techniques help in reducing the feature space while retaining important information to enhance model efficiency and performance. (b) You are building a handwriting recognition system to classify digits (0–9) from thousands of labelled images. To speed up processing, you plan to reduce the dimensionality of the dataset. Which technique would you prefer for dimensionality reduction in this case – **PCA** or **LDA**? Justify your choice.	CO2
	B	The following table shows the size of a house (in square feet) and its corresponding price (in ₹ lakhs): Size (sqft) | Price (₹ in lakhs) 1000 | 50 1500 | 65 2000 | 80 2500 | 95 3000 | 110 (a) Formulate a simple linear regression model for this data. (b) Calculate the slope and intercept of the best-fit line. (c) Use the model to predict the price of a house that is 2200 sqft in size.	CO2

CO1	Recall key concepts and terminology related to machine learning and its various techniques. (Remembering)
CO2	Explain the principles behind supervised, unsupervised, and reinforcement learning methods. (Understanding)
CO3	Apply appropriate machine learning algorithms to basic data-driven problems. (Applying)
CO4	Analyze the performance and behavior of machine learning models using standard evaluation metrics. (Analyzing)
CO5	Design and optimize machine learning solutions for real-world scenarios using advanced techniques and tools. (Evaluating)
CO6	Recall key concepts and terminology related to machine learning and its various techniques. (Remembering)